
Machine Translation

Philipp Koehn

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What is This?



- A class on machine translation
- Taught at Johns Hopkins University, Fall 2026
- Class web site: <http://www.mt-class.org/jhu/>
- Tuesdays and Thursdays, 1:30-2:45,
- Instructor: Philipp Koehn
- TA: Bismarck Odoom
- Grading
 - five programming assignments (10% each)
 - final project (30%)
 - in-class presentation: language in ten minutes (10%)
 - no exam, but ...

Attendance Required

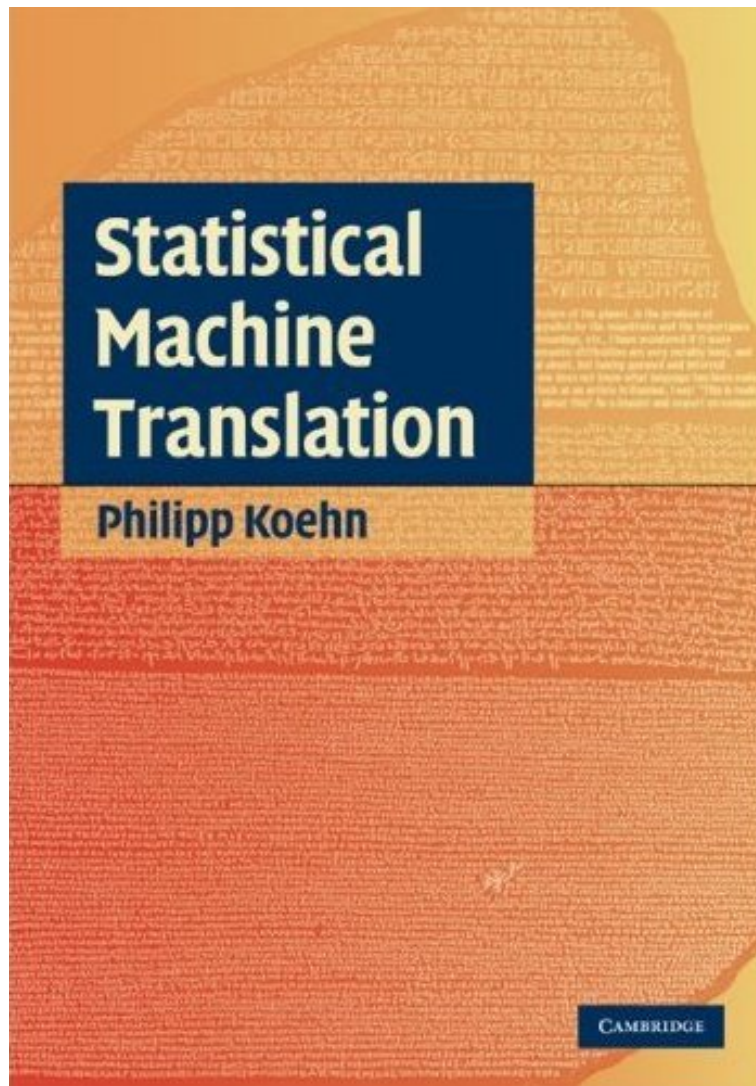
- 10% of the grade is attendance
 - Starting in the third week
 - Grading scheme
 - Starting point: 100 points
 - First lecture missed: -0 points
 - Second lecture missed: -1 points
 - Third lecture missed: -2 points
- $\text{score} = 100 - \text{missed}(\text{missed} - 1)$
- Possibly make-up exam to replace attendance grade?
(not decided yet, do not count on it)

Why Take This Class?



- Close look at an **artificial intelligence** problem
- Practical introduction to **natural language processing**
- Introduction to **deep learning** for structured prediction

Textbooks



some history

Warren Weaver on translation
as code breaking (1947):

*When I look at an article in Russian, I say:
"This is really written in English,
but it has been coded in some strange symbols.
I will now proceed to decode".*



Early Efforts and Disappointment



- Excited research in 1950s and 1960s

1954

Georgetown experiment

Machine could translate

250 words and

6 grammar rules



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- 1966 ALPAC report:
 - only \$20 million spent on translation in the US per year
 - no point in machine translation

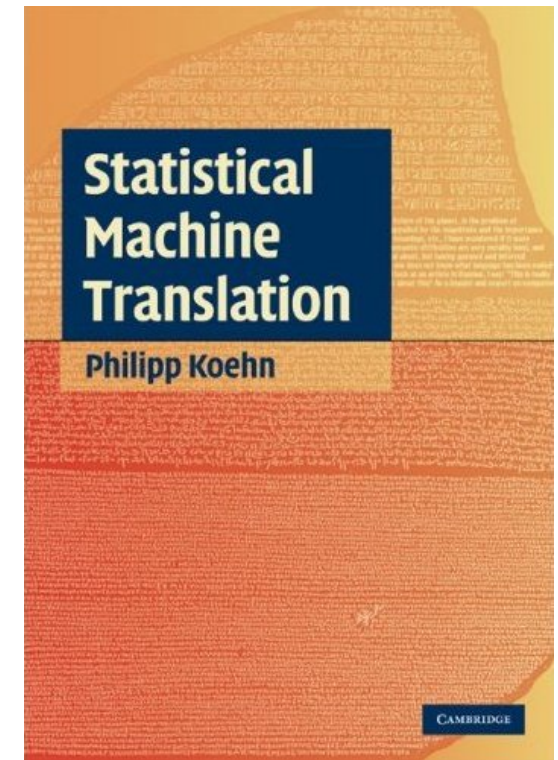
- Rule-based systems
 - build dictionaries
 - write transformation rules
 - refine, refine, refine
- Météo system for weather forecasts (1976)
- Systran (1968), Logos and Metal (1980s)

```
"have" :=  
  
if  
    subject (animate)  
    and object (owned-by-subject)  
then  
    translate to "kade... aahe"  
if  
    subject (animate)  
    and object (kinship-with-subject)  
then  
    translate to "laa... aahe"  
if  
    subject (inanimate)  
then  
    translate to "madhye...  
aahe"
```

Statistical Machine Translation

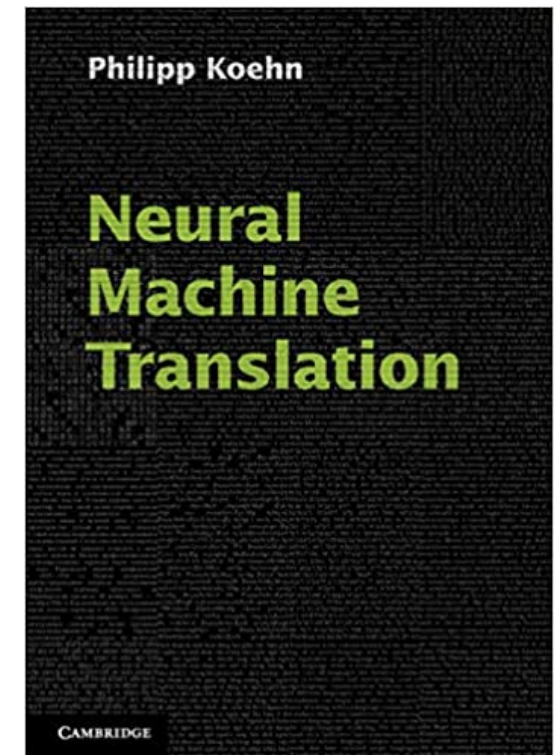


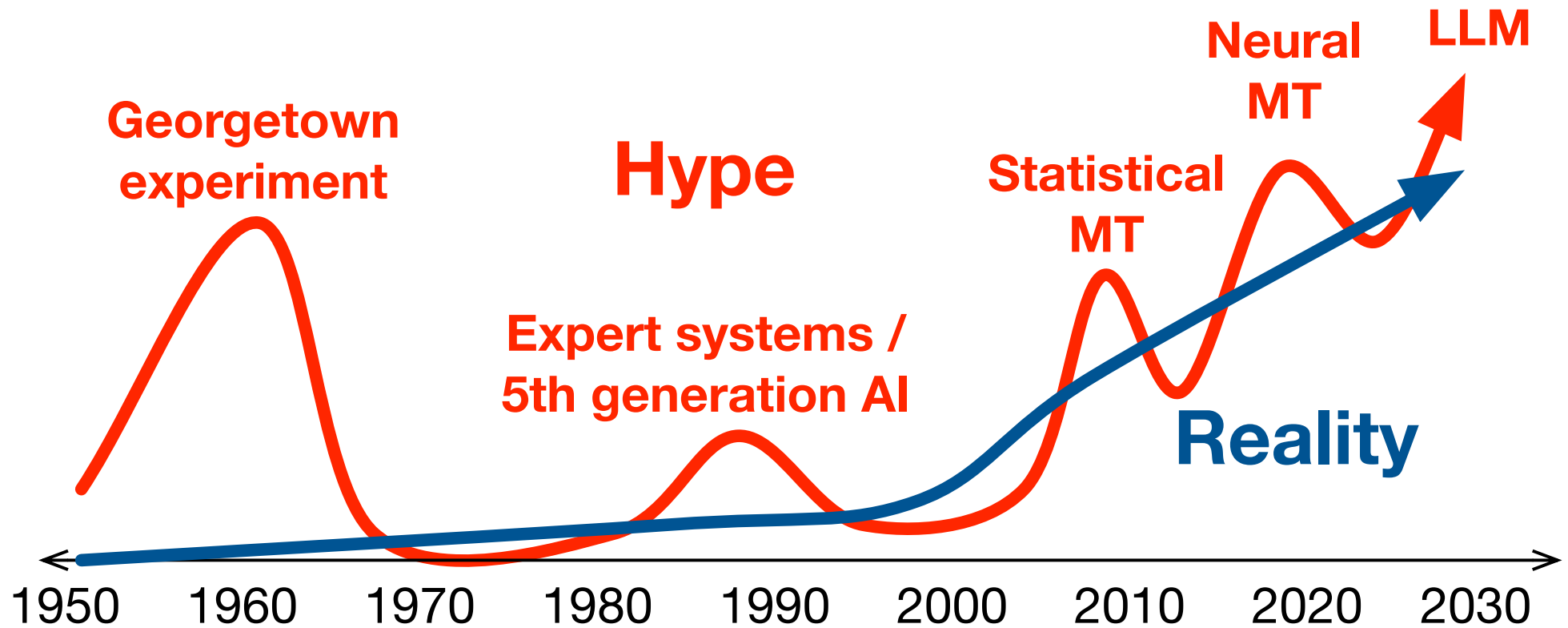
- 1980s: IBM
- 1990s: increased research
- Mid 2000s: Phrase-Based MT (Moses, Google)
- Around 2010: commercial viability



Neural Machine Translation

- Late 2000s: neural models for computer vision
- Since mid 2010s: neural models for machine translation
- 2016: neural machine translation the new state of the art
- 2024: increasing merge with large language models





how good is machine translation?

记者从环保部了解到,《水十条》要求今年年底前直辖市、省会城市、计划单列市建成区基本解决黑臭水体。截至目前,全国224个地级及以上城市共排查确认黑臭水体2082个,其中34.9%完成整治,28.4%正在整治,22.8%正在开展项目前期。



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Reporters learned from the Ministry of Environmental Protection, "Water 10" requirements before the end of this year before the municipality, the provincial capital city, plans to build a separate city to solve the basic black and black water. Up to now, the country's 224 prefecture-level and above cities were identified to confirm the black and white water 2082, of which 34.9% to complete the renovation, 28.4% is remediation, 22.8% is carrying out the project early.

A l'orée de ce débat télévisé inédit dans l'histoire de la Ve République, on attendait une forme de «Tous sur Macron» mais c'est la candidate du Front national qui s'est retrouvée au cœur des premières attaques de ses quatre adversaires d'un soir, favorisées par le premier thème abordé, les questions de société et donc de sécurité, d'immigration et de laïcité.

At the beginning of this televised debate, which was unheard of in the history of the Fifth Republic, a "Tous sur Macron" was expected, but it was the candidate of the National Front who found itself at the heart of the first attacks of its four Opponents of one evening, favored by the first theme tackled, the issues of society and thus security, immigration and secularism.

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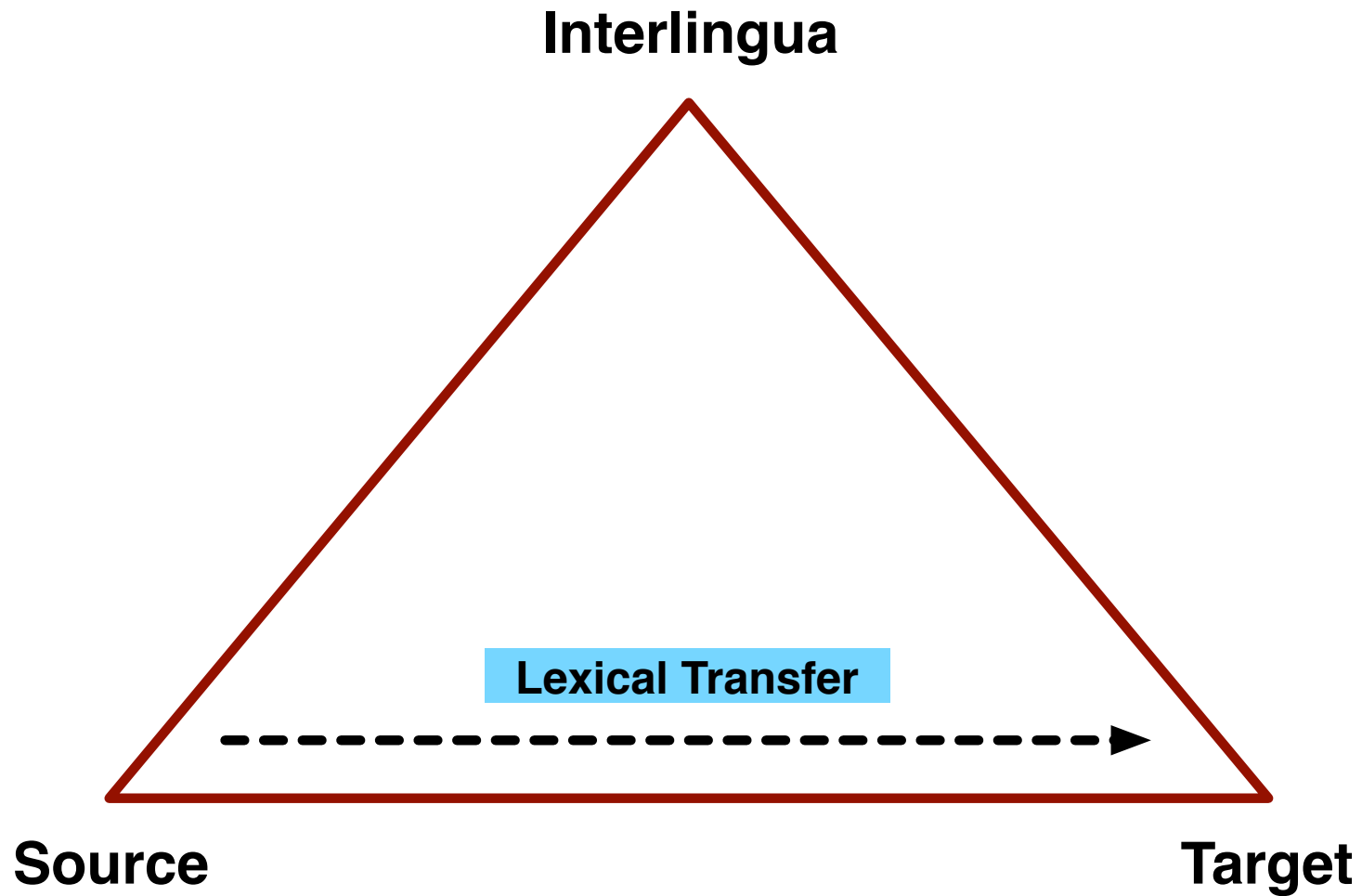
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Heading into this televised debate—unprecedented in the history of the Fifth Republic—a "gang up on Macron" scenario was anticipated; instead, however, it was the National Front candidate who found herself at the center of the initial attacks from her four opponents for the evening, a situation facilitated by the first topic addressed: social issues—and thus security, immigration, and secularism.

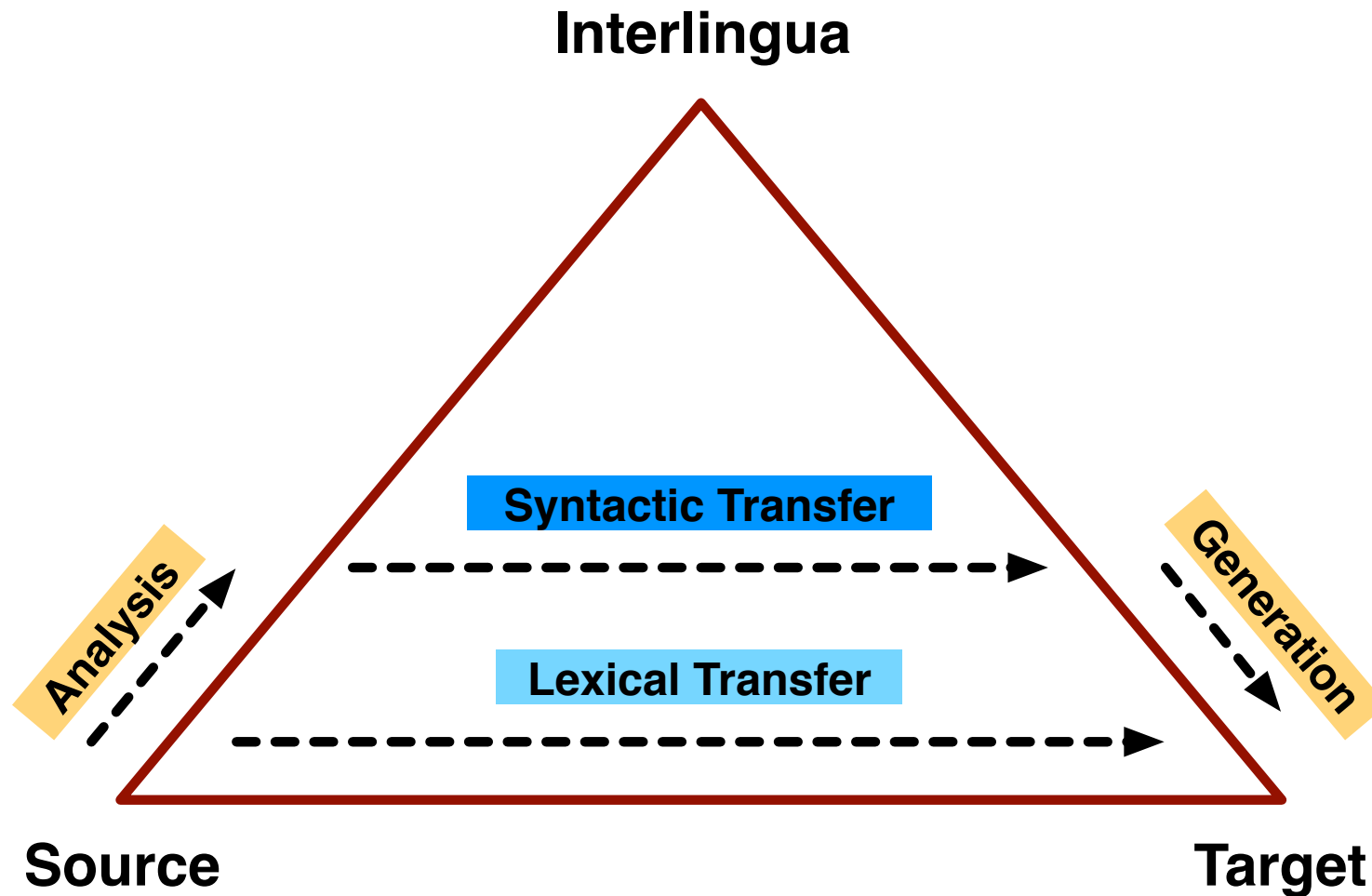
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The Ministry of Environmental Protection reports that the "Water Ten" action plan requires municipalities directly under the central government, provincial capitals, and cities with independent planning status to essentially eliminate black and odorous water bodies in their built-up areas by the end of this year. To date, a total of 2,082 such water bodies have been identified across 224 cities at or above the prefecture level; of these, 34.9% have undergone remediation, 28.4% are currently undergoing remediation, and 22.8% are in the preliminary project planning stage.

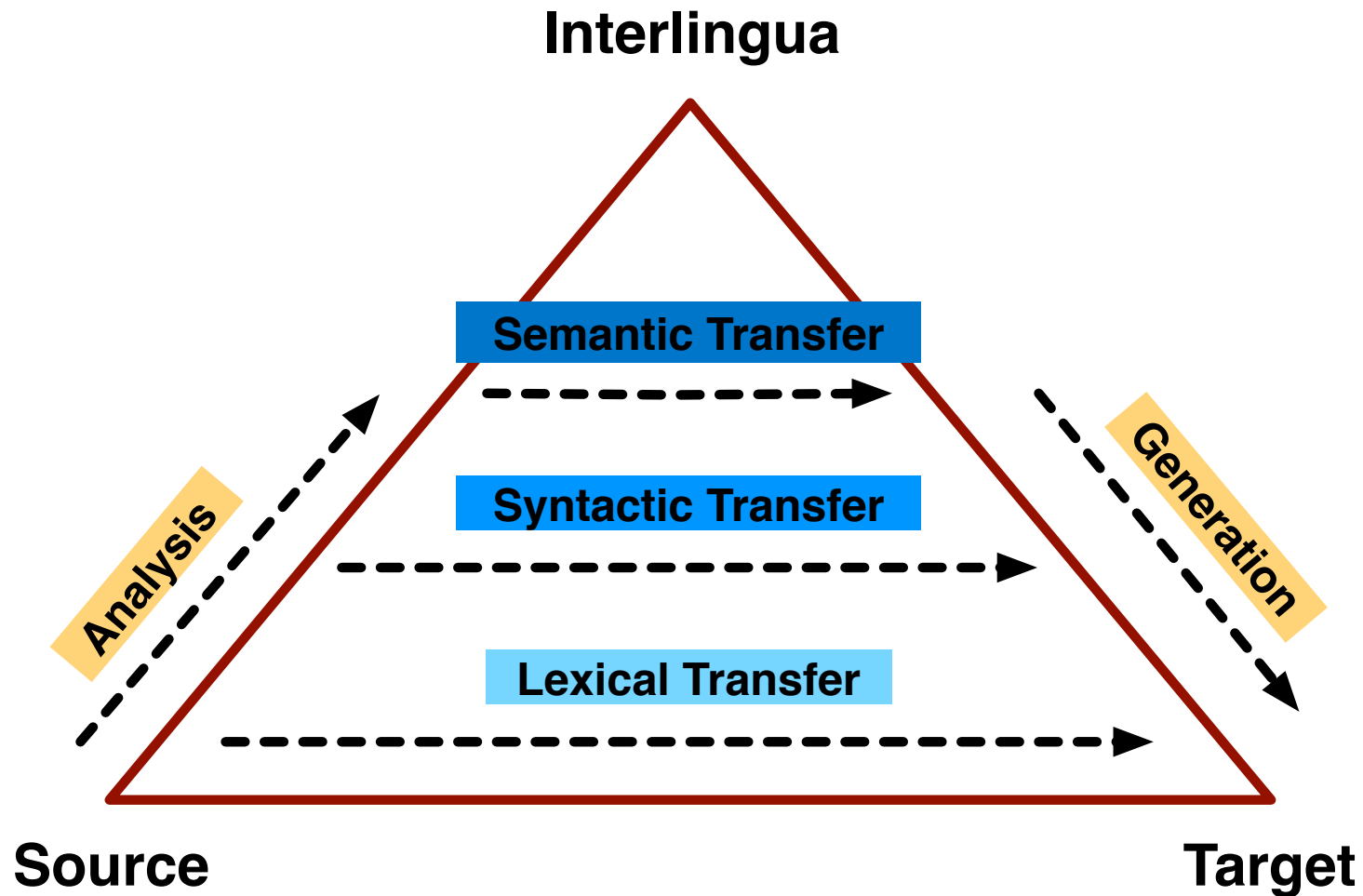
A Clear Plan



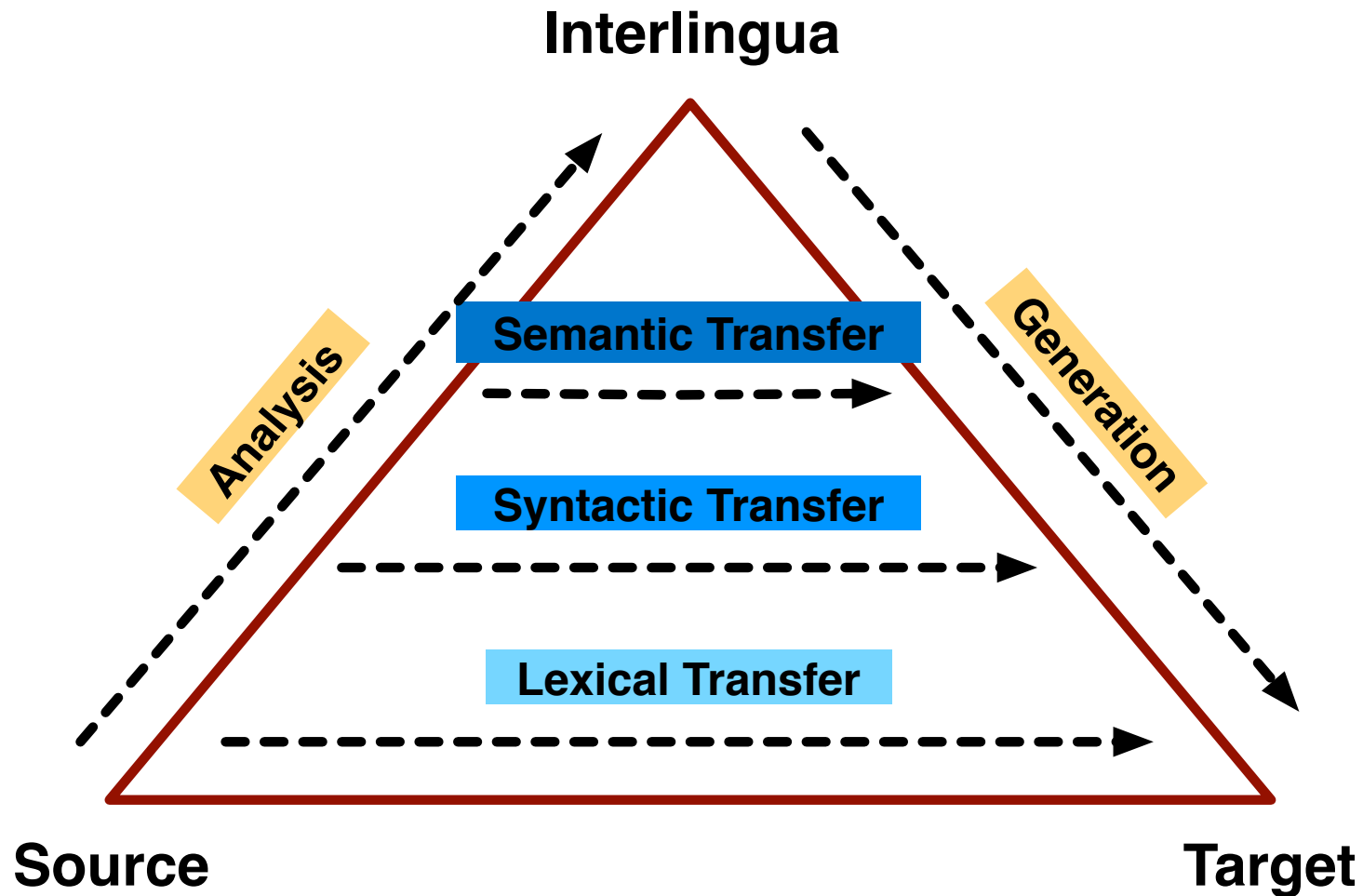
A Clear Plan



A Clear Plan



A Clear Plan



German documents

...Bank.....

.....Bank.....

.....Bank.....

.....Bank.....

English documents

...bank.....

.....bench.....

.....bank.....

.....bank.....

$$\Rightarrow p(\text{bank}|\text{Bank}) = 0.75, p(\text{bench}|\text{Bank}) = 0.25$$

why is that a good plan?

Word Translation Problems

- Words are ambiguous

He deposited money in a **bank** account
with a high **interest** rate.

Sitting on the **bank** of the Mississippi,
a passing ship piqued his **interest**.

- How do we find the right meaning, and thus translation?
- Context should be helpful

Syntactic Translation Problems

- Languages have different sentence structure

das	behaupten	sie	wenigstens
this	claim	they	at least
the		she	

- Convert from object-verb-subject (OVS) to subject-verb-object (SVO)
- Ambiguities can be resolved through syntactic analysis
 - the meaning **the** of **das** not possible (not a noun phrase)
 - the meaning **she** of **sie** not possible (subject-verb agreement)

- Pronominal anaphora

I saw the movie and **it** is good.

- How to translate **it** into German (or French)?
 - **it** refers to **movie**
 - **movie** translates to **Film**
 - **Film** has masculine gender
 - ergo: **it** must be translated into masculine pronoun **er**
- We are not handling this very well [Le Nagard and Koehn, 2010]

- Coreference

Whenever I visit my uncle and his daughters,
I can't decide who is my favorite **cousin**.

- How to translate **cousin** into German? Male or female?
- Complex inference required

- Discourse

Since you brought it up, I do not agree with you.

Since you brought it up, we have been working on it.

- How to translated *since*? Temporal or conditional?
- Analysis of discourse structure — a hard problem

- What is the best translation?

Sicherheit → security

Sicherheit → safety

Sicherheit → certainty

- What is the best translation?

Sicherheit → security 14,516

Sicherheit → safety 10,015

Sicherheit → certainty 334

- Counts in European Parliament corpus

- What is the best translation?

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- Phrasal rules

Sicherheitspolitik → security policy 1580

Sicherheitspolitik → safety policy 13

Sicherheitspolitik → certainty policy 0

Lebensmittelsicherheit → food security 51

Lebensmittelsicherheit → food safety 1084

Lebensmittelsicherheit → food certainty 0

Rechtssicherheit → legal security 156

Rechtssicherheit → legal safety 5

Rechtssicherheit → legal certainty 723

- What is most fluent?

a problem for translation

a problem of translation

a problem in translation

- What is most fluent?

a problem for translation 13,000

a problem of translation 61,600

a problem in translation 81,700

- Hits on Google

Learning from Data

- What is most fluent?

a problem for translation 13,000

a problem of translation 61,600

a problem in translation 81,700

a translation problem 235,000

- What is most fluent?

police disrupted the demonstration

police broke up the demonstration

police dispersed the demonstration

police ended the demonstration

police dissolved the demonstration

police stopped the demonstration

police suppressed the demonstration

police shut down the demonstration

- What is most fluent?

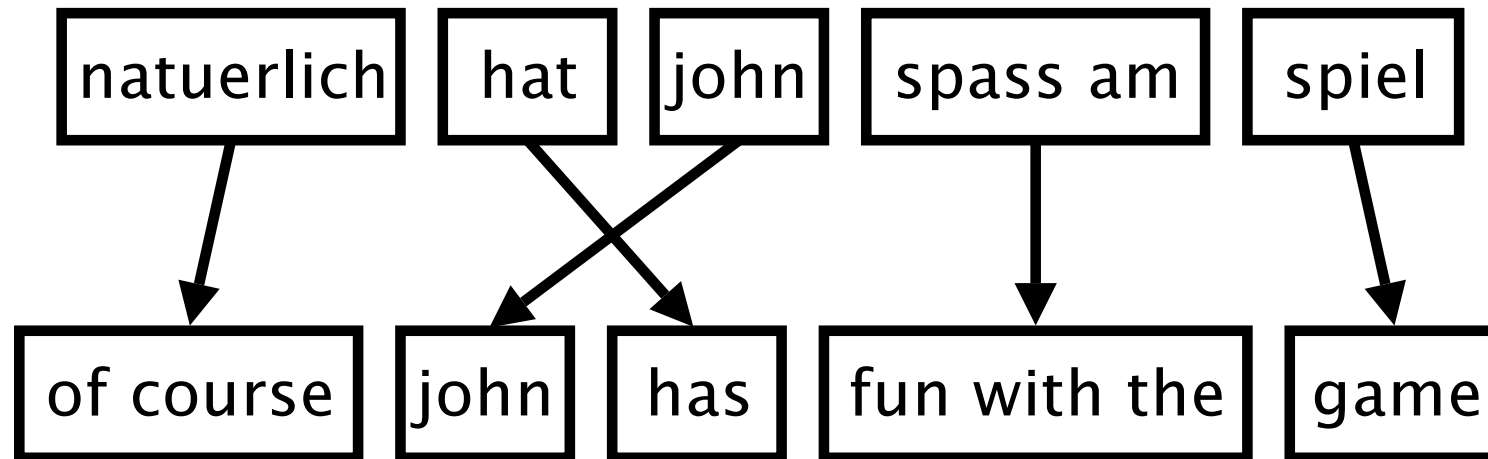
police disrupted the demonstration 2,140
police broke up the demonstration 66,600
police dispersed the demonstration 25,800
police ended the demonstration 762
police dissolved the demonstration 2,030
police stopped the demonstration 722,000
police suppressed the demonstration 1,400
police shut down the demonstration 2,040

where are we now?

Word Alignment

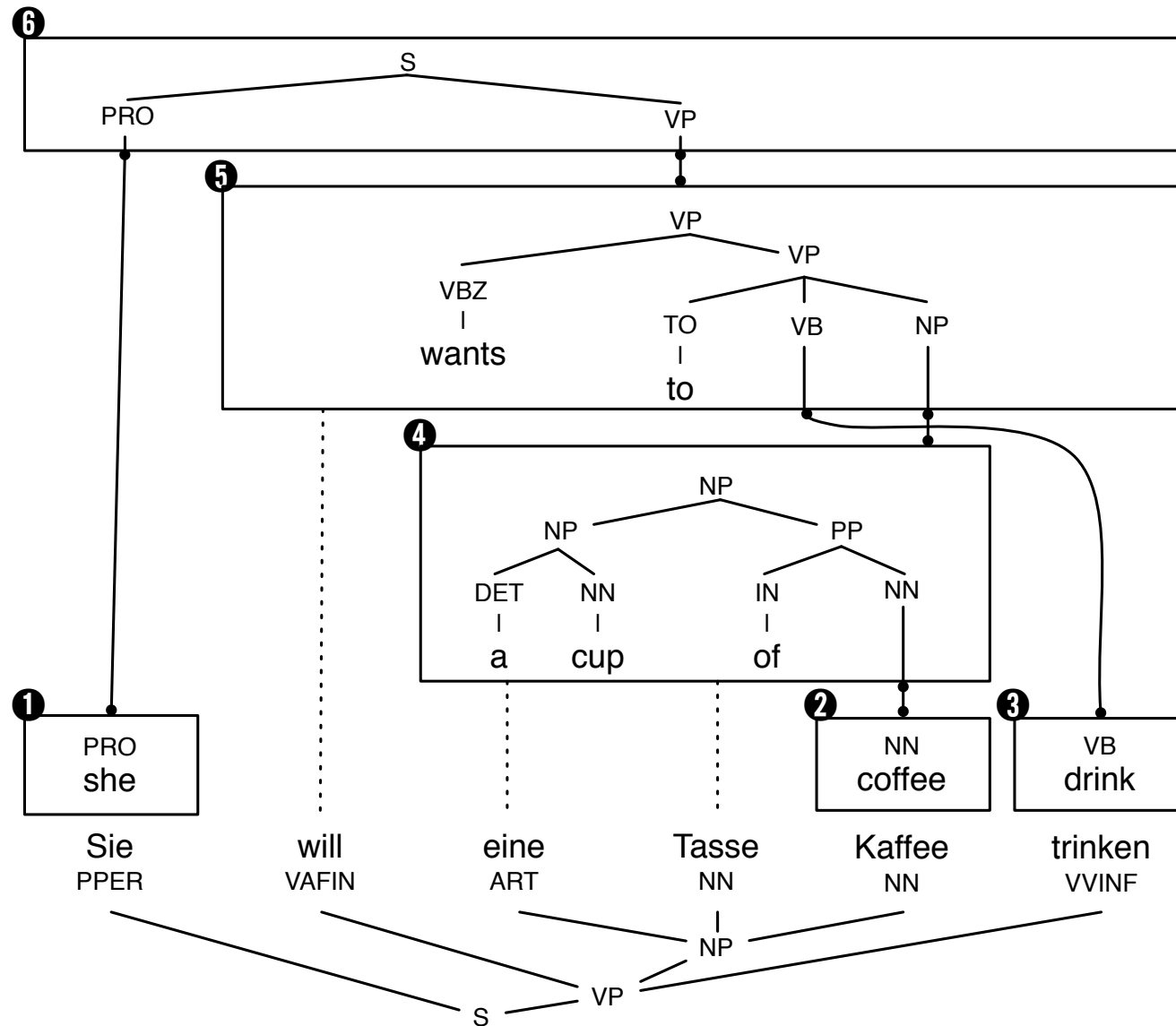
	michael	geht	davon	aus	,	dass	er	im	haus	bleibt
michael										
assumes										
that										
he										
will										
stay										
in										
the										
house										

Phrase-Based Model



- Foreign input is segmented in phrases
- Each phrase is translated into English
- Phrases are reordered
- Workhorse of today's statistical machine translation

Syntax-Based Translation

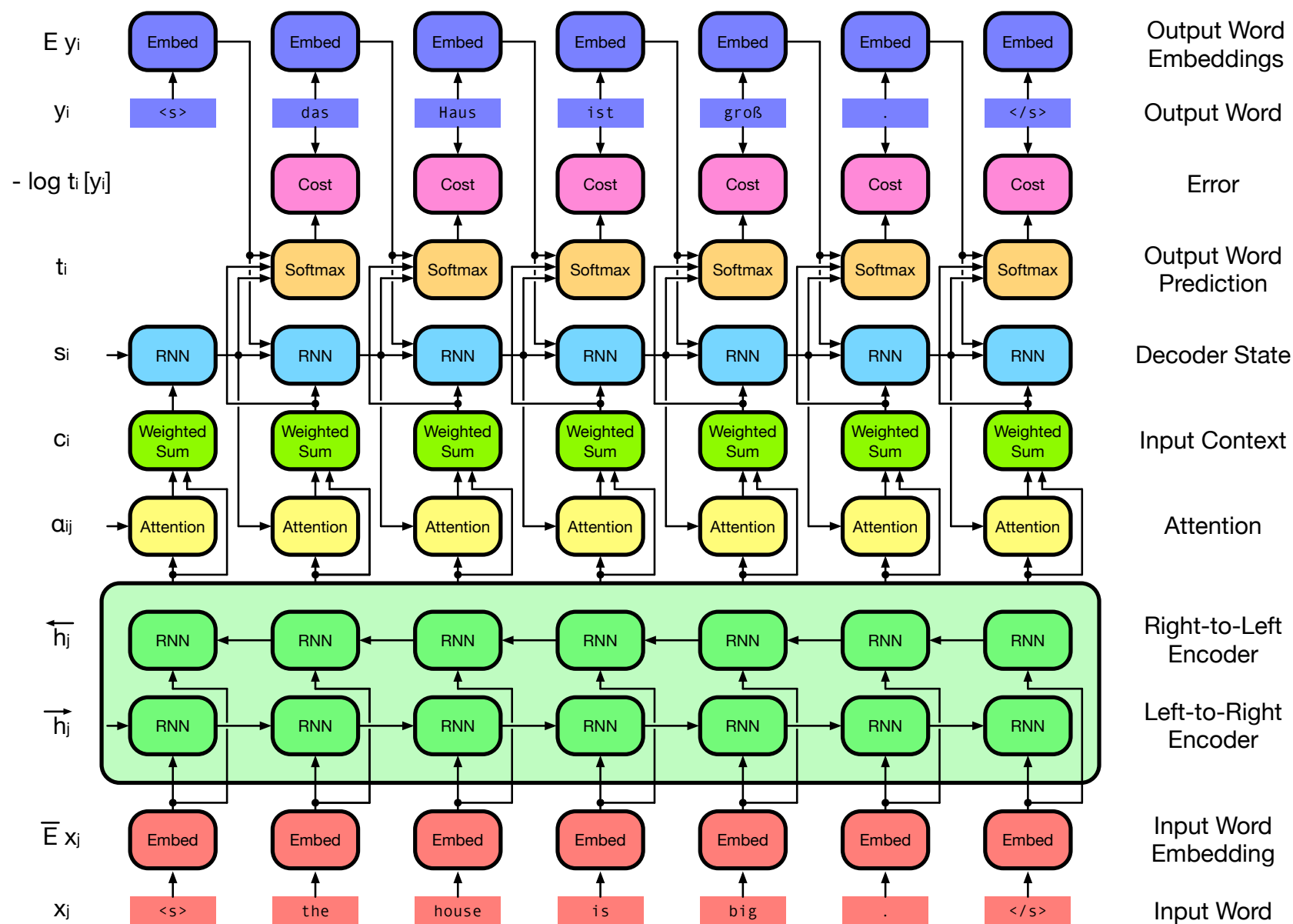


- Abstract meaning representation [Knight et al., ongoing]

```
(w / want-01
  :agent (b / boy)
  :theme (l / love
    :agent (g / girl)
    :patient b))
```

- Generalizes over equivalent syntactic constructs (e.g., active and passive)
- Defines semantic relationships
 - semantic roles
 - co-reference
 - discourse relations

Neural Model



Machine Translation and Language Models 43



- State-of-the-art model for machine translation: Transformer
- Transformer model was also adopted for language modeling



- State-of-the-art model for machine translation: Transformer
- Transformer model was also adopted for language modeling
- Currently, large language models being built by major IT companies (GPT4, Llama, Gemini, ...)
- Latest approach: fine-tuning large language models for machine translation

what is it good for?

what is it good *enough* for?

Why Machine Translation?

Assimilation — reader initiates translation, wants to know content

- user is tolerant of inferior quality
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- users can ask questions, when something is unclear
- chat room translations, hand-held devices
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Dissemination — publisher wants to make content available in other languages

- high demands for quality
- currently almost exclusively done by human translators

Problem: No Single Right Answer

这个 机场 的 安全 工作 由 以色列 方面 负责 .

Israeli officials are responsible for airport security.

Israel is in charge of the security at this airport.

The security work for this airport is the responsibility of the Israel government.

Israeli side was in charge of the security of this airport.

Israel is responsible for the airport's security.

Israel is responsible for safety work at this airport.

Israel presides over the security of the airport.

Israel took charge of the airport security.

The safety of this airport is taken charge of by Israel.

This airport's security is the responsibility of the Israeli security officials.

HTER	assessment
0%	
10%	publishable
20%	editable
30%	gistable
40%	triagable
50%	

(scale developed in preparation of DARPA GALE programme)

Applications

HTER	assessment	application examples
0%	publishable	Seamless bridging of language divide
10%		Automatic publication of official announcements
20%	editable	Increased productivity of human translators
30%		Access to official publications
40%	gistable	Multi-lingual communication (chat, social networks)
50%		Information gathering
	triagable	Trend spotting
		Identifying relevant documents

Current State of the Art

HTER	assessment	language pairs and domains
0%	publishable	French-English restricted domain
		French-English, Chinese-English news stories
10%	editable	French-English social media
		Swahili–English news stories
20%		
30%	gistable	Uyghur–English news stories
40%	triagable	Navajo-English
50%		

(informal rough estimates by presenter)

Thank You



questions?